Soln: a) For $x \in N$, $1 \leq x \leq 98$. Suppose the xth statement is True.

$\therefore$ "Exactly x of the statements in this list are false" is True.

$\therefore$ $(100-x)$ statements in the list are True.

$\therefore$ There exists a yth statement, where $x \neq y$ which is True

So, "Exactly y of the statements in this list are false" is True.

But, statement y is a contradiction to statement x.

$\boxed{\therefore \forall x \in N, 1 \leq x \leq 98, \text{ the } x\text{th statement must be False}}$

Suppose the 100th statement is True.

$\therefore$ "Exactly 100 of the statements in this list are false" is True.

So, the 100th statement by itself is both True and False at the same time

which is a contradiction to the definition of a Proposition.

$\therefore$ The 100th statement is False.

Suppose the 99th statement is True.

$\therefore$ "Exactly 99 of the statements in this list are false" is True.

We Know, Statements 1–98 and 100 are False. So the 99th statement is consistent with this fact.

b) Suppose the xth statement is true.

$\therefore$ "Atleast x of the statements in this list are false" is True,

$\therefore$ $\forall K \in N, 1 \leq K < x$, the Kth statement must also be true, $\because$ if atleast x statements are false, $K < x$, atleast K statements are naturally false.

$\therefore$ There are atleast x true statements, and atleast x false statements.

$$x + x = 100 \Rightarrow x = 50.$$ (Not rigorous)

$\therefore$ Statements 1–50 are True but Statements 51–100 are False.

c) $x + x = 99 \Rightarrow x = 49.5 \longrightarrow$ Such a situation cannot arise.